Advanced data visualization/: session 3

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setting the working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

importing libraries

```
library(ggplot2)
library(corrplot)
library(plotly)
library(ggiraph)
library(leaflet)
library(sf)
library(ggmap)
library(ggrepel)
library(ggridges)
library(patchwork)
library(dplyr)
library(GGally)
```

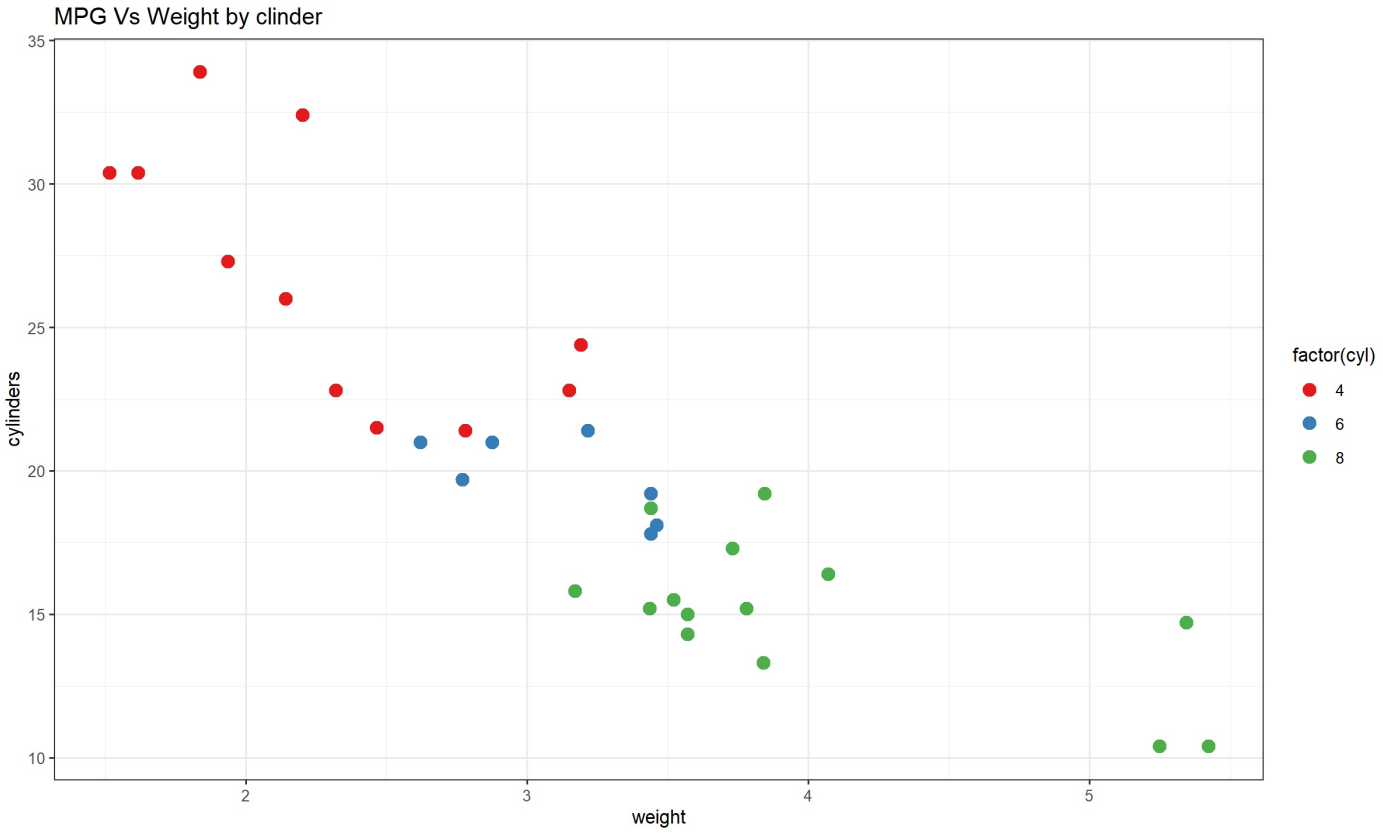
load data

```
data <- read.csv("GSSsubset.csv")
head(data)
```

```
  id    sex         degree   income  marital age height weight hrswrk
1  1   MALE       BACHELOR 60967.50 DIVORCED  53     72    190     60
2  2 FEMALE       BACHELOR 60967.50  MARRIED  26     60     97     40
3  4 FEMALE       BACHELOR 10161.25  MARRIED  56     68    160     20
4 14 FEMALE    HIGH SCHOOL 17551.25  MARRIED  40     65    156     37
5 16   MALE    HIGH SCHOOL 17551.25  MARRIED  56     NA    210      6
6 19   MALE LT HIGH SCHOOL 15703.75  MARRIED  51     68    170     NA
```

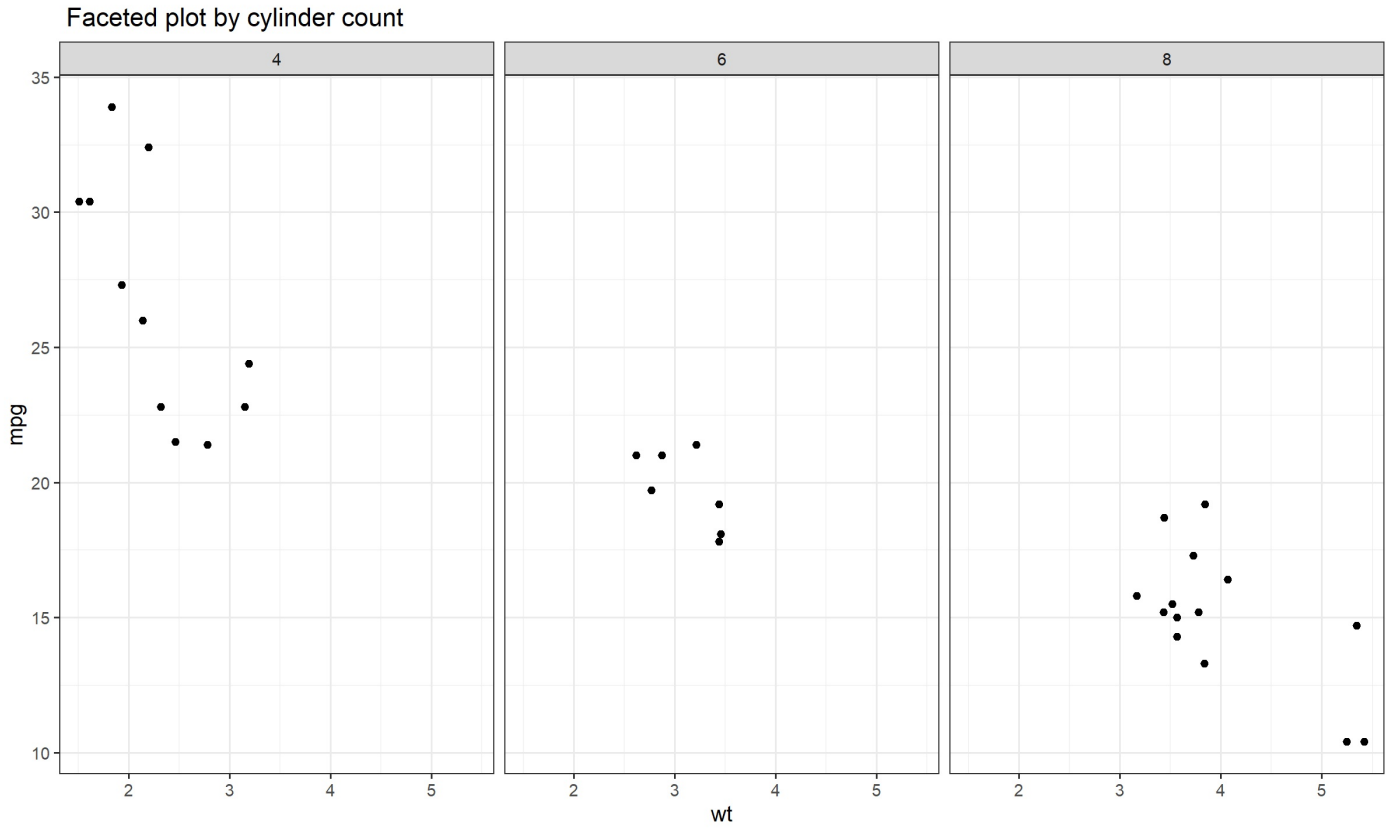
ggplot with custom themes and colour scales

```
p <- ggplot(mtcars, aes(x = wt, y = mpg, color = factor(cyl))) + geom_point(size = 3) +
    scale_color_brewer(palette = "Set1") + theme_bw(base_size = 10) + labs(title = "MPG Vs Weight by clinder",
    x = "weight", y = "cylinders")
p
```



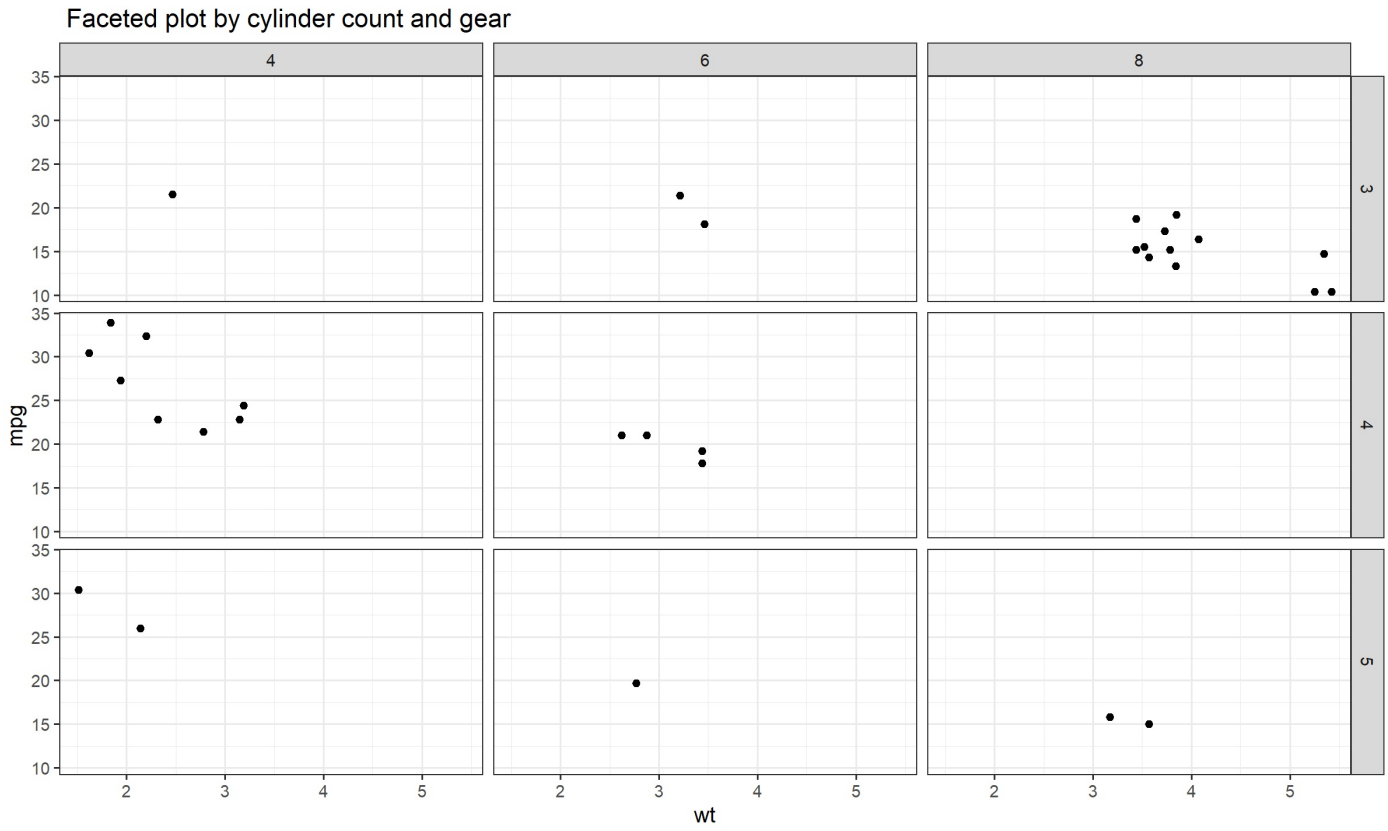
faceting and multi-panel plot

```
ggplot(mtcars, aes(x = wt, y = mpg)) + geom_point() + facet_wrap(~cyl) + theme_bw() +
    labs(title = " Faceted plot by cylinder count")
```



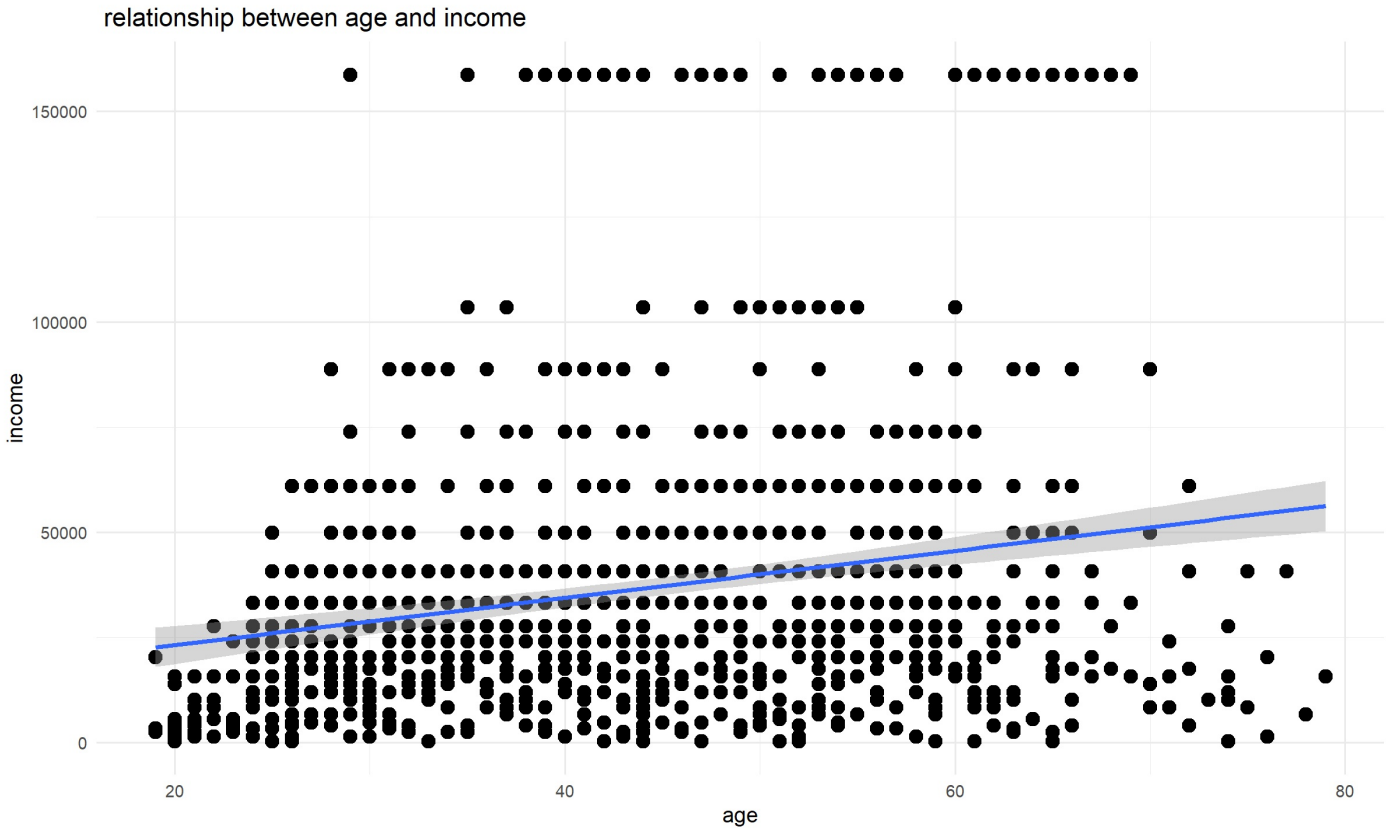
facet grid by cylinder and gears

```
ggplot(mtcars, aes(x = wt, y = mpg)) + geom_point() + facet_grid(gear ~ cyl) + theme_bw() +
    labs(title = " Faceted plot by cylinder count and gear")
```



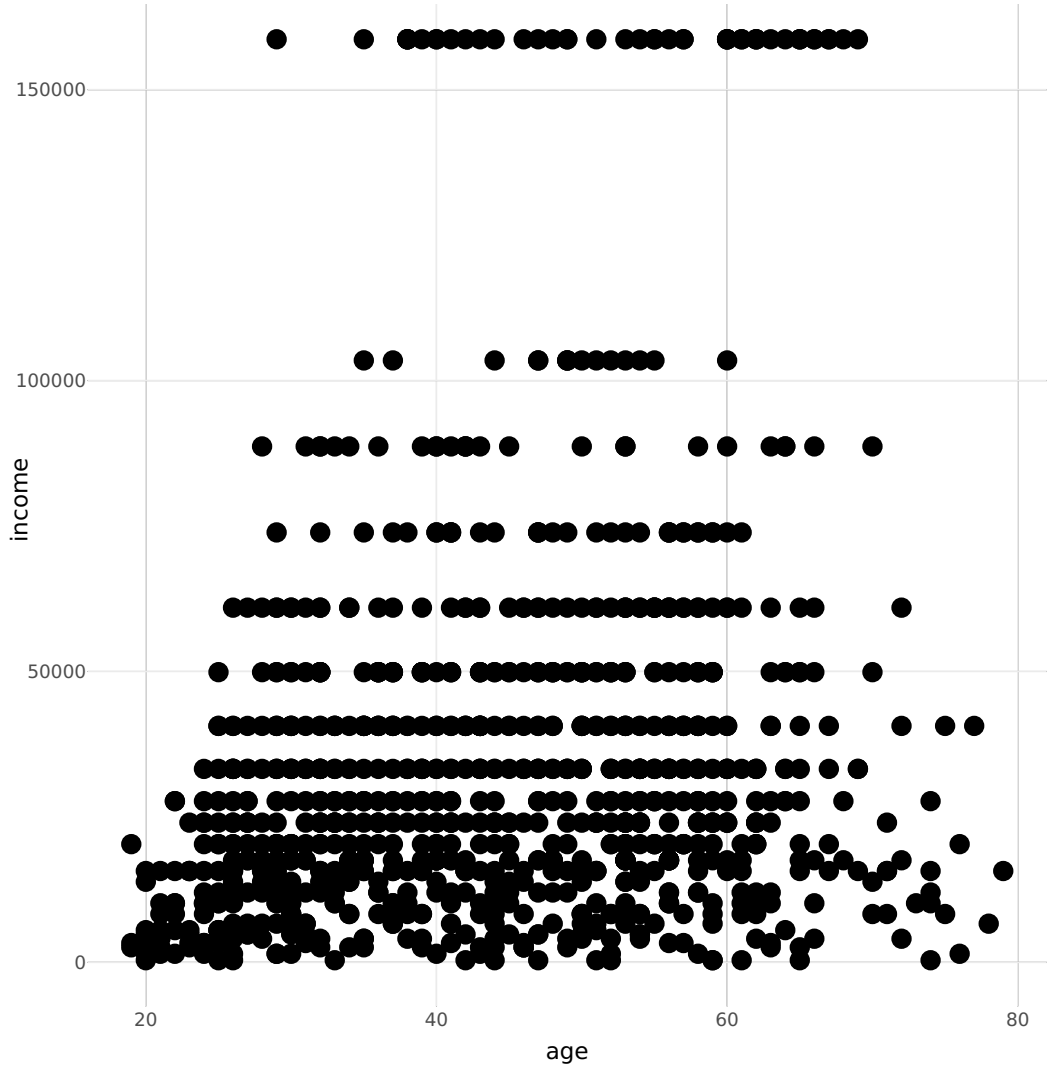
scatter plot + regression line

```
ggplot(data, aes(x = age, y = income)) + geom_point(size = 3) + geom_smooth(method = "lm",
    se = TRUE) + labs(title = " relationship between age and income", x = "age",
    y = "income") + theme_minimal()
```



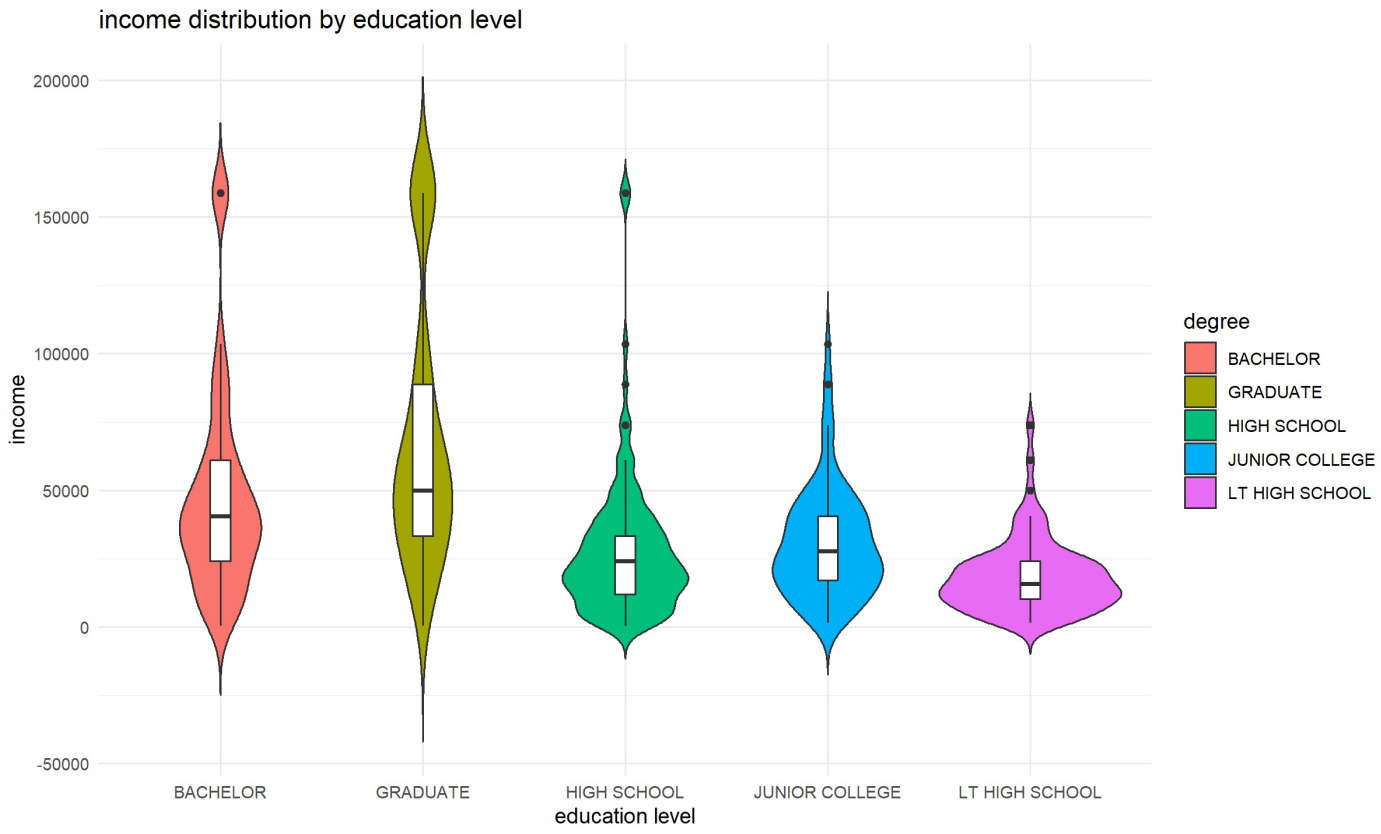
interactive scatter plot

```
p <- ggplot(data, aes(x = age, y = income, text = paste("sex:", sex, "<br>degree:",
    degree, "<br>income:", income, "<br>age:", age))) + geom_point(size = 3) + theme_minimal()
ggplotly(p, tooltip = "text")
```



violin plot

```
ggplot(data, aes(x = degree, y = income, fill = degree)) + geom_violin(trim = FALSE) +
    geom_boxplot(width = 0.1, fill = "white") + labs(title = "income distribution by education level",
    x = "education level", y = "income") + theme_minimal()
```

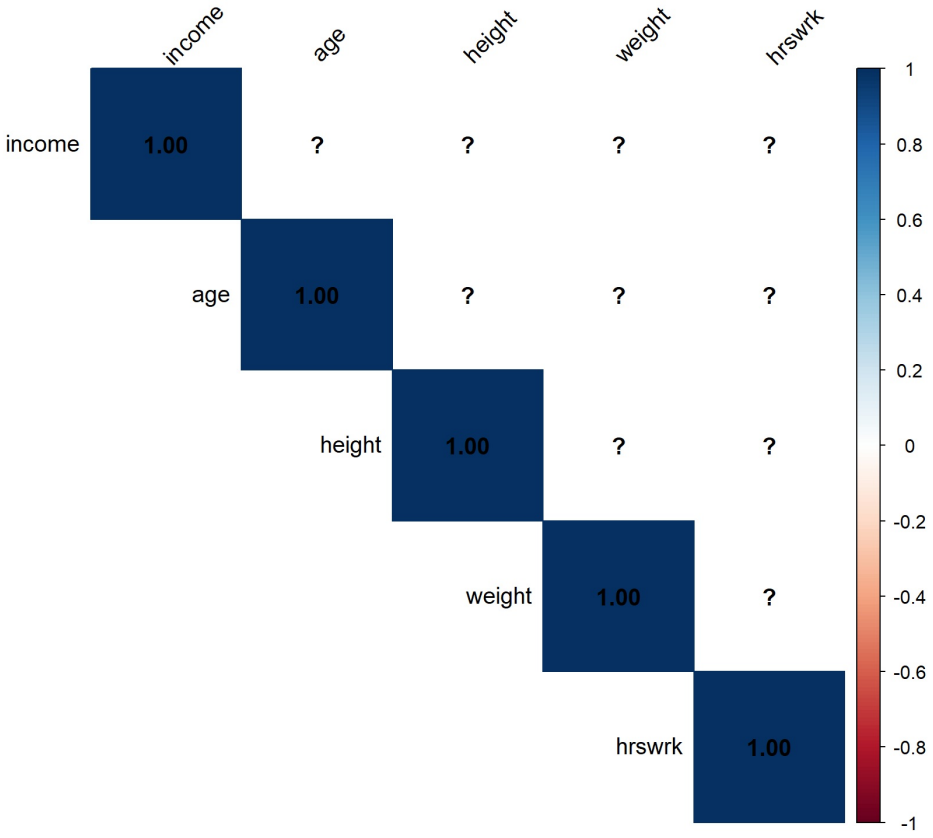


```
theme(legend.position = "none")
```

```
<theme> List of 1
 $ legend.position: chr "none"
 @ complete: logi FALSE
 @ validate: logi TRUE
```

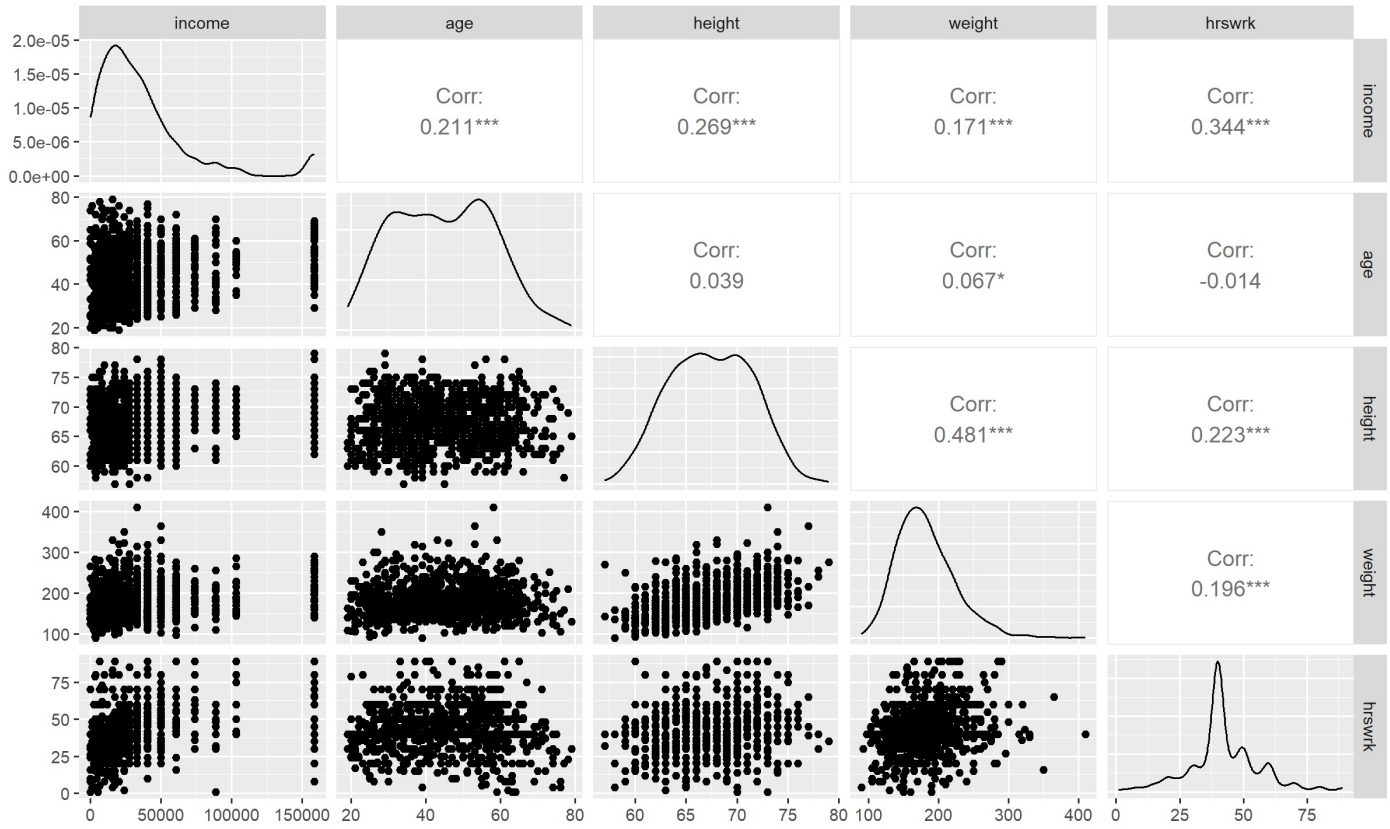
correlation heatmap

```
numeric_data <- data[, c("income", "age", "height", "weight", "hrswrk")]
cor_matrix <- cor(numeric_data)
corrplot(cor_matrix, method = "color", type = "upper", addCoef.col = "black", tl.col = "black",
    tl.srt = 45)
```



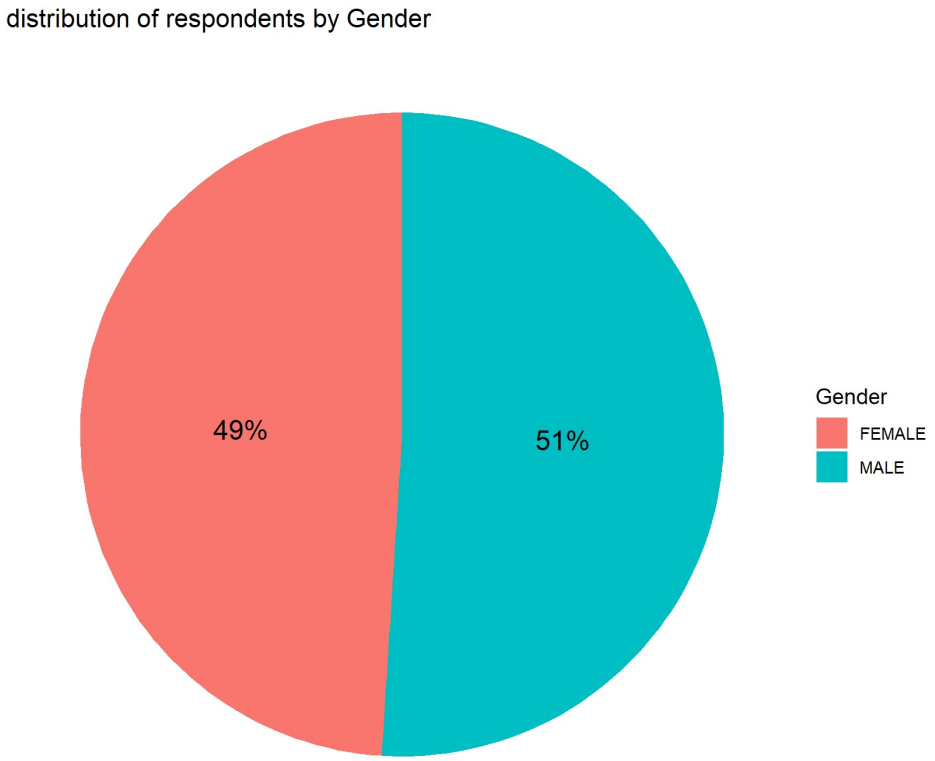
pairplot

```
ggpairs(data[, c("income", "age", "height", "weight", "hrswrk")])
```



pie chart count gender

```
set_data <- data %>%
    count(sex) %>%
    mutate(percentage = n/sum(n) * 100, label = paste0(round(percentage, 1), "%"))
ggplot(set_data, aes(x = "", y = n, fill = sex)) + geom_col(width = 1) + coord_polar(theta = "y") +
    geom_text(aes(label = label), position = position_stack(vjust = 0.5), size = 5) +
    labs(title = " distribution of respondents by Gender", fill = "Gender") + theme_void()
```



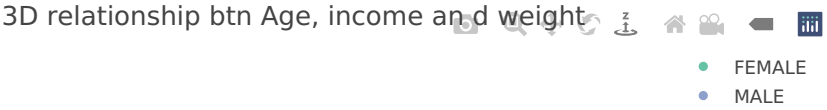
bubble chart

```
ggplot(data, aes(x = age, y = income, size = hrswrk, color = sex)) + geom_point(alpha = 0.5) +
    labs(title = "Age income and working hrs", x = "age", y = "income", size = "working hours",
        color = "sex") + theme_minimal(base_size = 20)
```



3D scatter plot

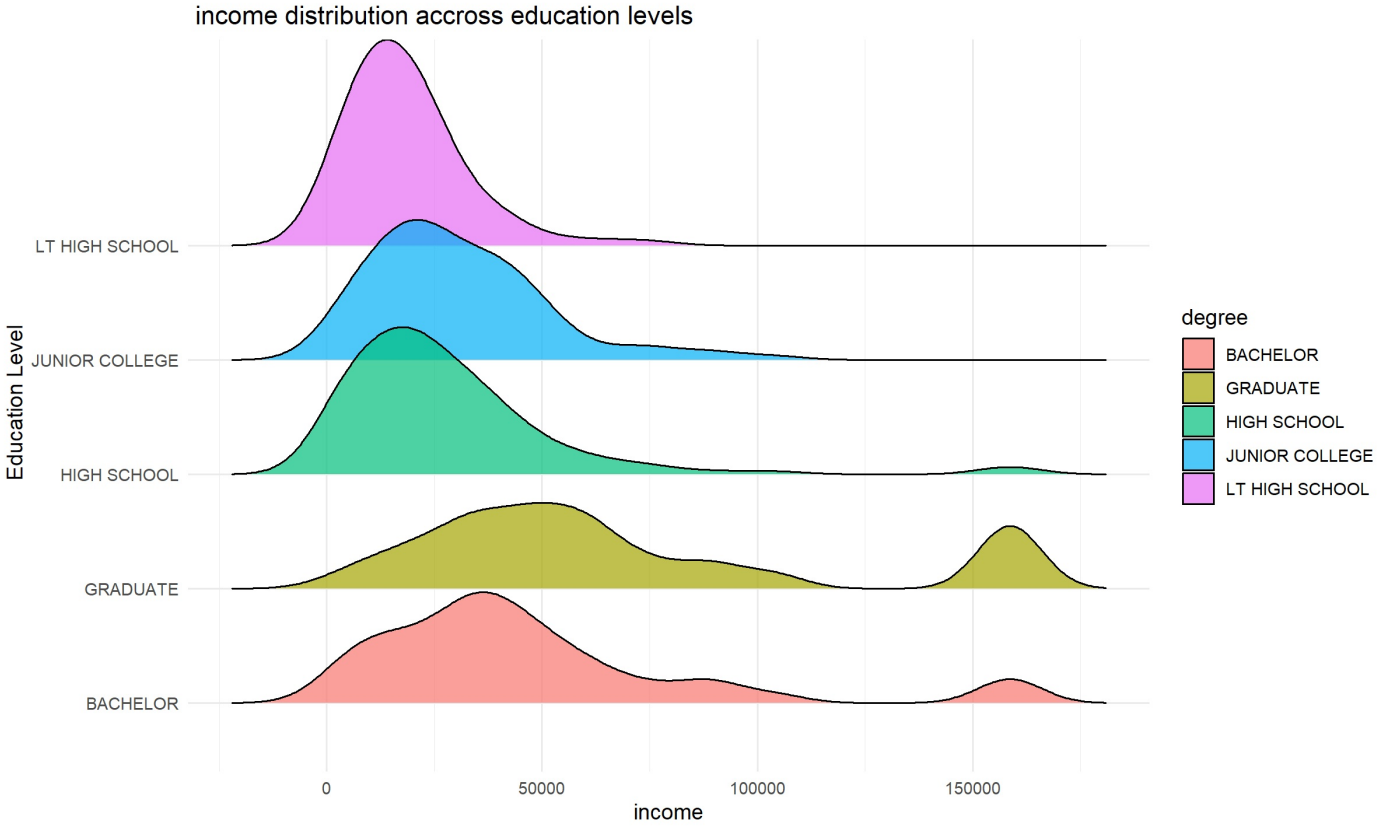
```
fig <- plot_ly(data, x = ~age, y = ~income, z = ~weight, color = ~sex, trpe = "scatter3d",
    mode = "markers", marker = list(size = 6))
fig <- fig %>%
    layout(title = "3D relationship btn Age, income an d weight", scene = list(xaxis = list(title = "Age"),
        yaxis = list(title = "Income"), zaxis = list(title = "weight")))
fig
```



WebGL is not supported by your browser - visit https://get.webgl.org for more info

ridge line/ distribution plot

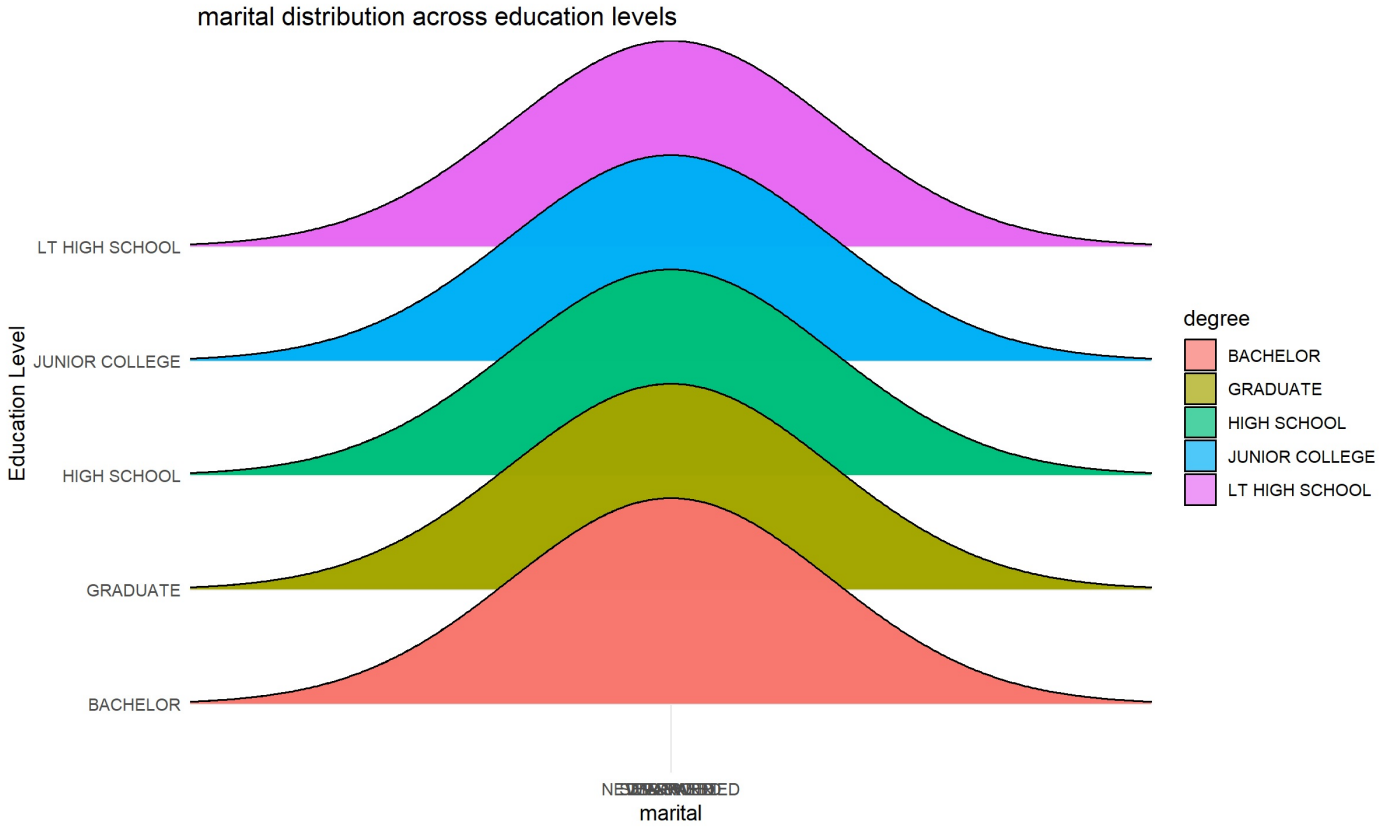
```
ggplot(data, aes(x = income, y = degree, fill = degree)) + geom_density_ridges(alpha = 0.7,
    bandwidth = 7430) + labs(title = " income distribution accross education levels",
    x = "income", y = " Education Level") + theme_minimal()
```




```
theme(legend.position = "none")
```

```
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 @ validate: logi TRUE
```

```
ggplot(data, aes(x = marital, y = degree, fill = degree)) + geom_density_ridges(alpha = 0.7,
    bandwidth = 7430) + labs(title = " marital distribution across education levels",
    x = "marital", y = " Education Level") + theme_minimal()
```




```
theme(legend.position = "none")
```

```
<theme> List of 1
 $ legend.position: chr "none"
 @ complete: logi FALSE
 @ validate: logi TRUE
```